

BUSA8090 - Data and Visualisation for Business Assignment 1

**David Jones Marketing Campaign Strategy:
Analytics and Database Design Report**

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Part A – Database Design and Development

1.1 Entity Identification

Entity Name	Entity Description
PLATFORM	Represents the social media platforms and their commission rate used by David Jones to promote the End of Financial Year (EOFY) sale, including YouTube, Instagram, Facebook, TikTok, and X.
ADVERTISEMENT	Represents the promotional advertisements or campaign posts created for the David Jones End of Financial Year sale. These ads are hosted on social media platforms and may generate customer engagement and orders.
CUSTOMER	Represents David Jones customers who place orders and make payments.
SHOP_ORDER	Represents customer orders placed during the End of Financial Year sale. It records order details such as date, customer, advertisement source, advertisement click that made an order and time of that click, shipping, and commission information.
SHIPPING	Represents the delivery details for customer orders made during the End of Financial Year sale, including recipient name, address, email, and phone number.
PRODUCT	Represents David Jones products promoted and sold during the End of Financial Year sale, including product category, price, and stock availability.
ORDER_PRODUCT	Represents the products included in each customer order. It acts as an associative entity between SHOP_ORDER and PRODUCT and records the product quantity and checkout price during the sale.

PAYMENT	Represents the information regarding payment transactions customers made for their David Jones End of Financial Year sale orders.
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1.2 Identify Attributes, Primary Keys and Foreign Keys

Entity	Attributes
PLATFORM	Platform_ID (Primary Key), Platform_Name, Platform_CommissionRate
ADVERTISEMENT	Ad_ID (Primary Key), Ad_Name, Ad_StartDate, Ad_EndDate, Platform_ID (Foreign Key referencing PLATFORM)
CUSTOMER	Customer_ID (Primary Key), Customer_FirstName, Customer_LastName, Customer_PhoneNo, Customer_Address, Customer_Email
SHOP_ORDER	Order_ID (Primary Key), Order_Date, Order_AdClickID, Order_AdClickTime, Order_CommissionRate, Customer_ID (Foreign Key referencing CUSTOMER), Shipping_ID (Foreign Key referencing SHIPPING), Ad_ID (Foreign Key referencing ADVERTISEMENT)
SHIPPING	Shipping_ID (Primary Key), Shipping_FirstName, Shipping_LastName, Shipping_Address, Shipping_Email, Shipping_PhoneNo
PRODUCT	Product_SKU (Primary Key), Product_Name, Product_Category, Product_Price, Product_QuantityInStock
ORDER_PRODUCT	Order_ID (Primary Key and Foreign Key referencing SHOP_ORDER), Product_SKU (Primary Key and Foreign Key referencing PRODUCT), Product_Quantity, Product_PriceAtCheckout

PAYMENT	Payment_ID (Primary Key), Payment_Method, Payment_Status, Customer_ID (Foreign Key referencing CUSTOMER), Order_ID (Foreign Key referencing SHOP_ORDER)
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1.3 Relationship Definition and Cardinality Analysis

PLATFORM hosts ADVERTISEMENT (1:M). One social media platform can host many advertisements, while each advertisement is hosted on one platform.

Business rule: Each advertisement must be assigned to exactly one social media platform, such as YouTube, Facebook, TikTok, Instagram, or X. A platform may have no advertisements or many advertisements, depending on whether it is used in a campaign.

ADVERTISEMENT generates SHOP_ORDER (1:M). One advertisement can generate many orders, while each order is linked to one advertisement.

Business rule: An advertisement may generate zero, one, or many customer orders. Each ad-driven order is linked to one advertisement so that campaign performance can be tracked. However, because some orders may not come from advertisements, the Ad_ID in SHOP_ORDER may be optional.

CUSTOMER places SHOP_ORDER (1:M). A customer may place many orders, while each order is placed by one customer.

Business rule: Each order must be placed by one customer. A customer can place multiple orders over time, but every order must be associated with a customer record for identification, contact, and order history purposes.

SHIPPING has SHOP_ORDER (1:M). One shipping record can be associated with many shop orders, while each shop order has one shipping record.

Business rule: Each order must have one shipping record that contains delivery details such as recipient name, address, email, and phone number. The same shipping record may be reused for multiple orders if the customer ships orders to the same recipient or address.

SHOP_ORDER contains PRODUCT (M:M). One order can include multiple products, and one product can appear in multiple orders. Because of this many-to-many relationship, the ORDER_PRODUCT entity is used as an associative entity to link SHOP_ORDER and PRODUCT and includes sale-specific details such as product quantity and product price at checkout.

Business rule: Each order must contain at least one product. A product may appear in many different orders. The ORDER_PRODUCT table records the products included in each order, the quantity purchased, and the product price at the time of checkout.

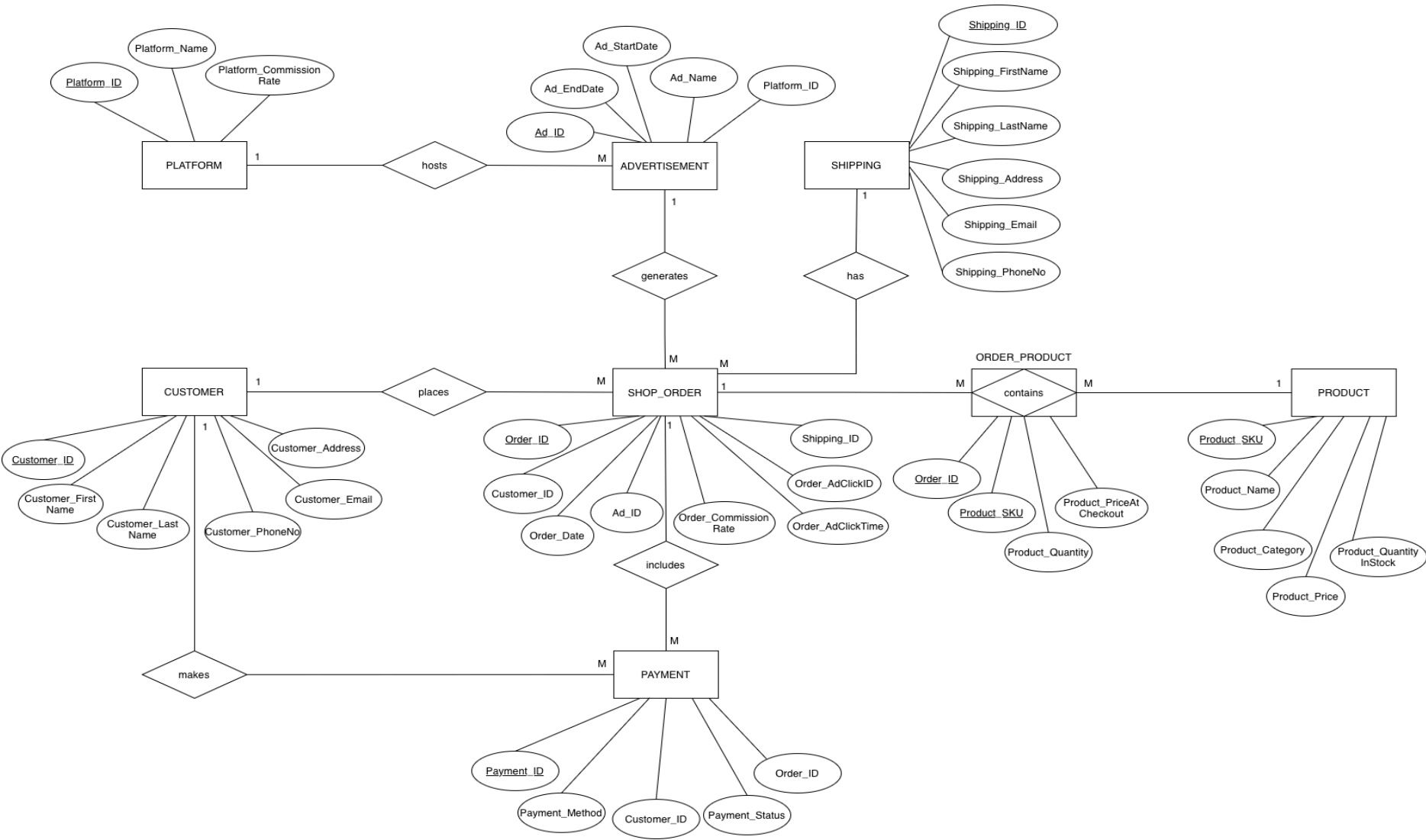
SHOP_ORDER includes PAYMENT (1:M). An order may involve many payment transactions. Each payment is linked to one order.

Business rule: Each payment must be linked to one specific order. An order may have one or more payment records, for example if there are failed and successful payment transactions. Payment records are used to track the payment method and payment status for each transaction.

CUSTOMER makes PAYMENT (1:M). One customer can make many payments across different orders, while each payment is made by one customer.

Business rule: Each payment must be made by one customer. A customer may make multiple payments over time, especially if they place multiple orders or make more than one payment attempt. This relationship helps track which customer is responsible for each payment transaction.

1.4 ERD diagram



Source: Author's work

1.5 Table implementation

```
CREATE TABLE PLATFORM (  
    Platform_ID      Integer      NOT NULL,  
    Platform_Name     Varchar(100) NOT NULL,  
    Platform_CommissionRate Decimal(5,3) NOT NULL,  
  
    CONSTRAINT PLATFORM_PK  
        PRIMARY KEY (Platform_ID)  
);
```

```
CREATE TABLE CUSTOMER (  
    Customer_ID      Integer      NOT NULL,  
    Customer_FirstName Varchar(100) NOT NULL,  
    Customer_LastName Varchar(100) NOT NULL,  
    Customer_PhoneNo  Varchar(20)  NOT NULL,  
    Customer_Address  Varchar(255) NOT NULL,  
    Customer_Email     Varchar(100) NOT NULL,  
  
    CONSTRAINT CUSTOMER_PK  
        PRIMARY KEY (Customer_ID)  
);
```

```
CREATE TABLE SHIPPING (  
    Shipping_ID      Integer      NOT NULL,  
    Shipping_FirstName  Varchar(100) NOT NULL,  
    Shipping_LastName   Varchar(100) NOT NULL,  
    Shipping_Address    Varchar(255) NOT NULL,  
    Shipping_Email      Varchar(100) NOT NULL,  
    Shipping_PhoneNo    Varchar(20) NOT NULL,  
  
    CONSTRAINT SHIPPING_PK  
        PRIMARY KEY (Shipping_ID)  
);
```

```
CREATE TABLE PRODUCT (  
    Product_SKU      Integer      NOT NULL,  
    Product_Name      Varchar(100) NOT NULL,  
    Product_Category   Varchar(100) NOT NULL,  
    Product_Price      Decimal(10,2) NOT NULL,  
    Product_QuantityInStock Integer NOT NULL,  
  
    CONSTRAINT PRODUCT_PK  
        PRIMARY KEY (Product_SKU)  
);
```

```
CREATE TABLE ADVERTISEMENT (
```

```
Ad_ID          Integer    NOT NULL,  
Platform_ID    Integer    NOT NULL,  
Ad_Name        Varchar(100) NOT NULL,  
Ad_StartDate   DateTime   NOT NULL,  
Ad_EndDate     DateTime   NULL,
```

```
CONSTRAINT AD_PK
```

```
    PRIMARY KEY (Ad_ID),
```

```
CONSTRAINT AD_PLATFORM_Relationship
```

```
    FOREIGN KEY (Platform_ID)
```

```
    REFERENCES PLATFORM (Platform_ID)
```

```
);
```

```
CREATE TABLE SHOP_ORDER (
```

```
    Order_ID      Integer    NOT NULL,  
    Customer_ID   Integer    NOT NULL,  
    Shipping_ID   Integer    NOT NULL,  
    Ad_ID         Integer    NULL,  
    Order_AdClickID Integer    NULL,  
    Order_AdClickTime DateTime  NULL,  
    Order_Date     DateTime   NOT NULL,  
    Order_CommissionRate Decimal(5,3) NULL,
```

CONSTRAINT SHOP_ORDER_PK

PRIMARY KEY (Order_ID),

CONSTRAINT ORDER_CUSTOMER_Relationship

FOREIGN KEY (Customer_ID)

REFERENCES CUSTOMER (Customer_ID),

CONSTRAINT ORDER_SHIPPING_Relationship

FOREIGN KEY (Shipping_ID)

REFERENCES SHIPPING (Shipping_ID),

CONSTRAINT ORDER_AD_Relationship

FOREIGN KEY (Ad_ID)

REFERENCES ADVERTISEMENT (Ad_ID)

);

CREATE TABLE ORDER_PRODUCT (

Order_ID Integer NOT NULL,

Product_SKU Integer NOT NULL,

Product_Quantity Integer NOT NULL,

Product_PriceAtCheckout Decimal(10,3) NOT NULL,

CONSTRAINT ORDER_PRODUCT_PK

PRIMARY KEY (Order_ID, Product_SKU),

CONSTRAINT ORDER_PRODUCT_ORDER_Relationship

FOREIGN KEY (Order_ID)

REFERENCES SHOP_ORDER (Order_ID),

CONSTRAINT ORDER_PRODUCT_PRODUCT_Relationship

FOREIGN KEY (Product_SKU)

REFERENCES PRODUCT (Product_SKU)

);

CREATE TABLE PAYMENT (

Payment_ID Integer NOT NULL,

Customer_ID Integer NOT NULL,

Order_ID Integer NOT NULL,

Payment_Method Varchar(50) NOT NULL,

Payment_Status Varchar(50) NOT NULL,

CONSTRAINT PAYMENT_PK

PRIMARY KEY (Payment_ID),

CONSTRAINT PAYMENT_CUSTOMER_Relationship

FOREIGN KEY (Customer_ID)

REFERENCES CUSTOMER (Customer_ID),

```
CONSTRAINT PAYMENT_ORDER_Relationship
    FOREIGN KEY (Order_ID)
    REFERENCES SHOP_ORDER (Order_ID)
);
```

1.6 Dummy data insertion

```
INSERT INTO PLATFORM (
    Platform_ID,
    Platform_Name,
    Platform_CommissionRate
)
```

```
VALUES
    (10001, 'Facebook', 0.05),
    (10002, 'TikTok', 0.06),
    (10003, 'Instagram', 0.055),
    (10004, 'X', 0.045),
    (10005, 'YouTube', 0.065);
```

```
INSERT INTO CUSTOMER (
    Customer_ID,
    Customer_FirstName,
    Customer_LastName,
    Customer_PhoneNo,
    Customer_Address,
```

Customer_Email

)

VALUES

(20001, 'Olivia', 'Bennett', '+61412034567', '14 Collins Street, Melbourne VIC 3000', 'olivia.bennett@gmail.com'),

(20002, 'Liam', 'Harris', '+61413045678', '22 George Street, Sydney NSW 2000', 'liam.harris@outlook.com'),

(20003, 'Charlotte', 'Wilson', '+61414056789', '8 Queen Street, Brisbane QLD 4000', 'charlotte.wilson@yahoo.com'),

(20004, 'Noah', 'Taylor', '+61415067890', '31 King William Street, Adelaide SA 5000', 'noah.taylor@gmail.com'),

(20005, 'Amelia', 'Martin', '+61416078901', '45 Murray Street, Perth WA 6000', 'amelia.martin@outlook.com'),

(20006, 'Jack', 'Thompson', '+61417089012', '19 Elizabeth Street, Hobart TAS 7000', 'jack.thompson@yahoo.com'),

(20007, 'Isla', 'Anderson', '+61418090123', '67 Northbourne Avenue, Canberra ACT 2601', 'isla.anderson@gmail.com'),

(20008, 'William', 'Clark', '+61419012345', '52 Smith Street, Darwin NT 0800', 'william.clark@outlook.com'),

(20009, 'Mia', 'Walker', '+61421023456', '11 Chapel Street, South Yarra VIC 3141', 'mia.walker@yahoo.com'),

(20010, 'Henry', 'Scott', '+61422034567', '90 Pitt Street, Sydney NSW 2000', 'henry.scott@gmail.com'),

(20011, 'Ava', 'Green', '+61423045678', '27 Ann Street, Brisbane QLD 4000', 'ava.green@outlook.com'),

(20012, 'Lucas', 'Hall', '+61424056789', '6 Rundle Mall, Adelaide SA 5000', 'lucas.hall@yahoo.com'),

(20013, 'Sophie', 'King', '+61425067890', '35 Hay Street, Perth WA 6000', 'sophie.king@gmail.com'),

(20014, 'Ethan', 'Wright', '+61426078901', '18 Macquarie Street, Hobart TAS 7000', 'ethan.wright@outlook.com'),

(20015, 'Grace', 'Mitchell', '+61427089012', '76 London Circuit, Canberra ACT 2601', 'grace.mitchell@yahoo.com'),

(20016, 'James', 'Roberts', '+61428090123', '42 Mitchell Street, Darwin NT 0800', 'james.roberts@gmail.com'),

(20017, 'Ella', 'Turner', '+61429012345', '21 Bourke Street, Melbourne VIC 3000', 'ella.turner@outlook.com'),

(20018, 'Benjamin', 'Phillips', '+61431023456', '59 Oxford Street, Sydney NSW 2021', 'benjamin.phillips@yahoo.com'),

(20019, 'Ruby', 'Campbell', '+61432034567', '17 Adelaide Street, Brisbane QLD 4000', 'ruby.campbell@gmail.com'),

(20020, 'Alexander', 'Parker', '+61433045678', '73 Grenfell Street, Adelaide SA 5000', 'alexander.parker@outlook.com'),

(20021, 'Chloe', 'Evans', '+61434056789', '29 St Georges Terrace, Perth WA 6000', 'chloe.evans@yahoo.com'),

(20022, 'Mason', 'Edwards', '+61435067890', '12 Liverpool Street, Hobart TAS 7000', 'mason.edwards@gmail.com'),

(20023, 'Zoe', 'Collins', '+61436078901', '84 Bunda Street, Canberra ACT 2601', 'zoe.collins@outlook.com'),

(20024, 'Thomas', 'Stewart', '+61437089012', '5 Cavenagh Street, Darwin NT 0800', 'thomas.stewart@yahoo.com'),

(20025, 'Lily', 'Morris', '+61438090123', '101 Swanston Street, Melbourne VIC 3000', 'lily.morris@gmail.com');

INSERT INTO SHIPPING (

```
Shipping_ID,  
Shipping_FirstName,  
Shipping_LastName,  
Shipping_Address,  
Shipping_Email,  
Shipping_PhoneNo  
)  
VALUES  
  
    (30001, 'Olivia', 'Bennett', '14 Collins Street, Melbourne VIC 3000',  
    'olivia.bennett@gmail.com', '+61412034567'),  
  
    (30002, 'Liam', 'Harris', '22 George Street, Sydney NSW 2000',  
    'liam.harris@outlook.com', '+61413045678'),  
  
    (30003, 'Charlotte', 'Wilson', '8 Queen Street, Brisbane QLD 4000',  
    'charlotte.wilson@yahoo.com', '+61414056789'),  
  
    (30004, 'Noah', 'Taylor', '31 King William Street, Adelaide SA 5000',  
    'noah.taylor@gmail.com', '+61415067890'),  
  
    (30005, 'Amelia', 'Martin', '45 Murray Street, Perth WA 6000',  
    'amelia.martin@outlook.com', '+61416078901'),  
  
    (30006, 'Jack', 'Thompson', '19 Elizabeth Street, Hobart TAS 7000',  
    'jack.thompson@yahoo.com', '+61417089012'),  
  
    (30007, 'Isla', 'Anderson', '67 Northbourne Avenue, Canberra ACT 2601',  
    'isla.anderson@gmail.com', '+61418090123'),  
  
    (30008, 'William', 'Clark', '52 Smith Street, Darwin NT 0800',  
    'william.clark@outlook.com', '+61419012345'),  
  
    (30009, 'Mia', 'Walker', '11 Chapel Street, South Yarra VIC 3141',  
    'mia.walker@yahoo.com', '+61421023456'),
```

(30010, 'Henry', 'Scott', '90 Pitt Street, Sydney NSW 2000', 'henry.scott@gmail.com', '+61422034567'),

(30011, 'Ava', 'Green', '27 Ann Street, Brisbane QLD 4000', 'ava.green@outlook.com', '+61423045678'),

(30012, 'Lucas', 'Hall', '6 Rundle Mall, Adelaide SA 5000', 'lucas.hall@yahoo.com', '+61424056789'),

(30013, 'Sophie', 'King', '35 Hay Street, Perth WA 6000', 'sophie.king@gmail.com', '+61425067890'),

(30014, 'Ethan', 'Wright', '18 Macquarie Street, Hobart TAS 7000', 'ethan.wright@outlook.com', '+61426078901'),

(30015, 'Grace', 'Mitchell', '76 London Circuit, Canberra ACT 2601', 'grace.mitchell@yahoo.com', '+61427089012'),

(30016, 'James', 'Roberts', '42 Mitchell Street, Darwin NT 0800', 'james.roberts@gmail.com', '+61428090123'),

(30017, 'Ella', 'Turner', '21 Bourke Street, Melbourne VIC 3000', 'ella.turner@outlook.com', '+61429012345'),

(30018, 'Benjamin', 'Phillips', '59 Oxford Street, Sydney NSW 2021', 'benjamin.phillips@yahoo.com', '+61431023456'),

(30019, 'Ruby', 'Campbell', '17 Adelaide Street, Brisbane QLD 4000', 'ruby.campbell@gmail.com', '+61432034567'),

(30020, 'Alexander', 'Parker', '73 Grenfell Street, Adelaide SA 5000', 'alexander.parker@outlook.com', '+61433045678'),

(30021, 'Chloe', 'Evans', '29 St Georges Terrace, Perth WA 6000', 'chloe.evans@yahoo.com', '+61434056789'),

(30022, 'Mason', 'Edwards', '12 Liverpool Street, Hobart TAS 7000', 'mason.edwards@gmail.com', '+61435067890'),

(30023, 'Zoe', 'Collins', '84 Bunda Street, Canberra ACT 2601', 'zoe.collins@outlook.com', '+61436078901'),

(30024, 'Thomas', 'Stewart', '5 Cavenagh Street, Darwin NT 0800',
'thomas.stewart@yahoo.com', '+61437089012'),

(30025, 'Lily', 'Morris', '101 Swanston Street, Melbourne VIC 3000',
'lily.morris@gmail.com', '+61438090123');

INSERT INTO PRODUCT (

Product_SKU,

Product_Name,

Product_Category,

Product_Price,

Product_QuantityInStock

)

VALUES

(40001, 'Country Road Linen Shirt', 'Women Fashion', 129.95, 85),

(40002, 'Trenery Wool Blend Coat', 'Women Fashion', 349.00, 42),

(40003, 'Seed Heritage Knit Jumper', 'Women Fashion', 99.95, 70),

(40004, 'Calvin Klein Slim Fit Shirt', 'Men Fashion', 119.00, 64),

(40005, 'Politix Tailored Blazer', 'Men Fashion', 299.00, 35),

(40006, 'R.M. Williams Leather Belt', 'Accessories', 149.00, 58),

(40007, 'Mimco Crossbody Bag', 'Accessories', 249.00, 31),

(40008, 'Oroton Leather Wallet', 'Accessories', 179.00, 44),

(40009, 'Nike Air Max Sneakers', 'Footwear', 189.99, 76),

(40010, 'Adidas Ultraboost Shoes', 'Footwear', 219.99, 53),

(40011, 'Clinique Moisture Surge Cream', 'Beauty', 74.00, 120),

(40012, 'Estee Lauder Night Repair Serum', 'Beauty', 165.00, 90),

(40013, 'Dior Sauvage Eau de Parfum', 'Fragrance', 185.00, 68),
(40014, 'Chanel Coco Mademoiselle', 'Fragrance', 210.00, 52),
(40015, 'Dyson Supersonic Hair Dryer', 'Electrical', 599.00, 27),
(40016, 'Breville Barista Express Coffee Machine', 'Home Appliances', 749.00, 21),
(40017, 'Sheridan Cotton Quilt Cover Set', 'Home', 229.00, 39),
(40018, 'Salt&Pepper Dinner Set', 'Home', 139.00, 62),
(40019, 'Scanpan Classic Frypan', 'Kitchen', 159.00, 47),
(40020, 'Tefal Non-Stick Cookware Set', 'Kitchen', 299.00, 33),
(40021, 'Apple AirPods Pro', 'Technology', 399.00, 48),
(40022, 'Samsung Galaxy Watch', 'Technology', 449.00, 29),
(40023, 'Lego Creator Expert Set', 'Toys', 179.99, 41),
(40024, 'Bonds Kids Pyjama Set', 'Kids', 39.95, 110),
(40025, 'Polo Ralph Lauren Kids Polo Shirt', 'Kids', 89.95, 73);

INSERT INTO ADVERTISEMENT (

Ad_ID,

Platform_ID,

Ad_Name,

Ad_StartDate,

Ad_EndDate

)

VALUES

(50001, 10001, 'David Jones EOFY Fashion Clearance Facebook Ad',
'2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50002, 10001, 'David Jones Beauty Bonus Facebook Ad', '2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50003, 10001, 'David Jones Homewares EOFY Facebook Ad', '2026-06-02 08:00:00', '2026-06-30 23:59:59'),

(50004, 10001, 'David Jones Designer Accessories Facebook Ad', '2026-06-03 08:00:00', '2026-06-30 23:59:59'),

(50005, 10001, 'David Jones EOFY Free Delivery Facebook Ad', '2026-06-04 08:00:00', '2026-06-30 23:59:59'),

(50006, 10002, 'David Jones EOFY TikTok Fashion Reel', '2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50007, 10002, 'David Jones Beauty Haul TikTok Ad', '2026-06-02 08:00:00', '2026-06-30 23:59:59'),

(50008, 10002, 'David Jones Winter Wardrobe TikTok Ad', '2026-06-03 08:00:00', '2026-06-30 23:59:59'),

(50009, 10002, 'David Jones Luxury Gifts TikTok Ad', '2026-06-04 08:00:00', '2026-06-30 23:59:59'),

(50010, 10002, 'David Jones EOFY Final Days TikTok Ad', '2026-06-10 08:00:00', '2026-06-30 23:59:59'),

(50011, 10003, 'David Jones Instagram EOFY Style Edit', '2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50012, 10003, 'David Jones Instagram Beauty Sale Story', '2026-06-02 08:00:00', '2026-06-30 23:59:59'),

(50013, 10003, 'David Jones Instagram Home Sale Carousel', '2026-06-03 08:00:00', '2026-06-30 23:59:59'),

(50014, 10003, 'David Jones Instagram Footwear Sale', '2026-06-04 08:00:00', '2026-06-30 23:59:59'),

(50015, 10003, 'David Jones Instagram EOFY Luxe Picks', '2026-06-05 08:00:00', '2026-06-30 23:59:59'),

(50016, 10004, 'David Jones X EOFY Flash Sale Post', '2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50017, 10004, 'David Jones X Beauty Deal Post', '2026-06-02 08:00:00', '2026-06-30 23:59:59'),

(50018, 10004, 'David Jones X Home Essentials Sale', '2026-06-03 08:00:00', '2026-06-30 23:59:59'),

(50019, 10004, 'David Jones X Final Week EOFY Reminder', '2026-06-10 08:00:00', '2026-06-30 23:59:59'),

(50020, 10004, 'David Jones X Luxury Accessories Offer', '2026-06-11 08:00:00', '2026-06-30 23:59:59'),

(50021, 10005, 'David Jones YouTube EOFY Sale Pre-Roll', '2026-06-01 08:00:00', '2026-06-30 23:59:59'),

(50022, 10005, 'David Jones YouTube Beauty EOFY Campaign', '2026-06-02 08:00:00', '2026-06-30 23:59:59'),

(50023, 10005, 'David Jones YouTube Home Sale Campaign', '2026-06-03 08:00:00', '2026-06-30 23:59:59'),

(50024, 10005, 'David Jones YouTube Technology Deals', '2026-06-04 08:00:00', '2026-06-30 23:59:59'),

(50025, 10005, 'David Jones YouTube EOFY Last Chance', '2026-06-10 08:00:00', '2026-06-30 23:59:59');

INSERT INTO SHOP_ORDER (

Order_ID,

Customer_ID,

Shipping_ID,

Ad_ID,

Order_Date,

```
Order_CommissionRate,  
Order_AdClickID,  
Order_AdClickTime  
)  
VALUES  
  
(60001, 20001, 30001, NULL, '2026-06-05 09:30:02', NULL, NULL, NULL),  
  
(60002, 20002, 30002, 50006, '2026-06-05 19:50:00', 0.06, 70001, '2026-06-05  
19:35:00'),  
  
(60003, 20003, 30003, 50011, '2026-06-06 12:05:44', 0.055, 70002, '2026-06-06  
11:48:52'),  
  
(60004, 20004, 30004, NULL, '2026-06-06 13:15:08', NULL, NULL, NULL),  
  
(60005, 20005, 30005, 50021, '2026-06-06 18:30:11', 0.065, 70003, '2026-06-06  
18:12:03'),  
  
(60006, 20006, 30006, NULL, '2026-06-07 10:45:00', NULL, NULL, NULL),  
  
(60007, 20007, 30007, 50002, '2026-06-07 20:10:01', 0.06, 70004, '2026-06-07  
19:52:00'),  
  
(60008, 20008, 30008, 50012, '2026-06-08 14:25:09', 0.055, 70005, '2026-06-08  
14:02:00'),  
  
(60009, 20009, 30009, 50017, '2026-06-08 15:40:08', 0.045, 70006, '2026-06-08  
15:21:00'),  
  
(60010, 20010, 30010, 50022, '2026-06-08 19:15:10', 0.065, 70007, '2026-06-08  
18:57:00'),  
  
(60011, 20001, 30001, NULL, '2026-06-09 09:55:00', NULL, NULL, NULL),  
  
(60012, 20002, 30002, 50024, '2026-06-10 04:30:45', 0.05, 70008, '2026-06-10  
02:11:02'),  
  
(60013, 20011, 30011, 50013, '2026-06-10 13:20:27', 0.055, 70009, '2026-06-10  
12:58:04'),
```

(60014, 20012, 30012, 50018, '2026-06-10 14:50:00', 0.045, 70010, '2026-06-10 14:29:10'),

(60015, 20013, 30013, NULL, '2026-06-10 19:45:25', NULL, NULL, NULL),

(60016, 20014, 30014, 50003, '2026-06-11 11:05:22', 0.05, 70011, '2026-06-11 10:46:03'),

(60017, 20015, 30015, 50009, '2026-06-11 18:40:08', 0.06, 70012, '2026-06-11 18:16:05'),

(60018, 20016, 30016, NULL, '2026-06-11 20:35:44', NULL, NULL, NULL),

(60019, 20017, 30017, 50019, '2026-06-12 12:10:32', 0.045, 70013, '2026-06-12 11:47:22'),

(60020, 20018, 30018, 50024, '2026-06-12 13:11:06', 0.065, 70014, '2026-06-12 13:09:48'),

(60021, 20019, 30019, 50005, '2026-06-13 10:15:42', 0.05, 70015, '2026-06-13 09:58:09'),

(60022, 20020, 30020, 50010, '2026-06-13 05:00:03', 0.06, 70016, '2026-06-13 04:53:10'),

(60023, 20021, 30021, 50015, '2026-06-14 16:20:08', 0.055, 70017, '2026-06-14 15:57:11'),

(60024, 20022, 30022, NULL, '2026-06-14 17:45:43', NULL, NULL, NULL),

(60025, 20023, 30023, 50025, '2026-06-14 21:40:24', 0.065, 70018, '2026-06-14 21:18:04'),

(60026, 20024, 30024, 50005, '2026-06-15 09:20:45', 0.05, 70019, '2026-06-15 09:03:03'),

(60027, 20025, 30025, 50010, '2026-06-15 22:50:32', 0.06, 70020, '2026-06-15 22:27:02'),

(60028, 20003, 30003, NULL, '2026-06-16 12:35:17', NULL, NULL, NULL),

(60029, 20010, 30010, NULL, '2026-06-17 10:05:24', NULL, NULL, NULL);

```
INSERT INTO ORDER_PRODUCT (
```

```
    Order_ID,
```

```
    Product_SKU,
```

```
    Product_Quantity,
```

```
    Product_PriceAtCheckout
```

```
)
```

```
VALUES
```

```
    (60001, 40001, 2, 89.95),
```

```
    (60002, 40009, 1, 149.99),
```

```
    (60003, 40011, 2, 59.00),
```

```
    (60004, 40018, 1, 99.00),
```

```
    (60005, 40021, 1, 349.00),
```

```
    (60006, 40012, 1, 129.00),
```

```
    (60007, 40003, 3, 69.95),
```

```
    (60008, 40007, 1, 199.00),
```

```
    (60009, 40019, 2, 119.00),
```

```
    (60010, 40013, 1, 149.00),
```

```
    (60011, 40002, 1, 279.00),
```

```
    (60012, 40010, 1, 179.99),
```

```
    (60013, 40017, 2, 169.00),
```

```
    (60014, 40020, 1, 239.00),
```

```
    (60015, 40016, 1, 649.00),
```

```
    (60016, 40015, 1, 529.00),
```

(60017, 40004, 2, 89.00),
(60018, 40014, 1, 179.00),
(60019, 40006, 1, 119.00),
(60020, 40022, 1, 399.00),
(60021, 40008, 2, 139.00),
(60022, 40005, 1, 249.00),
(60023, 40023, 1, 139.99),
(60024, 40024, 4, 29.95),
(60025, 40025, 2, 69.95),
(60026, 40001, 1, 89.95),
(60027, 40012, 2, 129.00),
(60027, 40020, 1, 239.00),
(60028, 40001, 2, 89.95),
(60029, 40007, 1, 199.00);

INSERT INTO PAYMENT (

Payment_ID,

Customer_ID,

Order_ID,

Payment_Method,

Payment_Status

)

VALUES

(70001, 20001, 60001, 'Credit Card', 'Paid'),

(70002, 20002, 60002, 'PayPal', 'Paid'),
(70003, 20003, 60003, 'Afterpay', 'Paid'),
(70004, 20004, 60004, 'Debit Card', 'Paid'),
(70005, 20005, 60005, 'Credit Card', 'Paid'),
(70006, 20006, 60006, 'PayPal', 'Paid'),
(70007, 20007, 60007, 'Afterpay', 'Pending'),
(70008, 20008, 60008, 'Credit Card', 'Paid'),
(70009, 20009, 60009, 'Debit Card', 'Paid'),
(70010, 20010, 60010, 'PayPal', 'Paid'),
(70011, 20001, 60011, 'Credit Card', 'Paid'),
(70012, 20002, 60012, 'Afterpay', 'Paid'),
(70013, 20011, 60013, 'Debit Card', 'Paid'),
(70014, 20012, 60014, 'Credit Card', 'Refunded'),
(70015, 20013, 60015, 'PayPal', 'Paid'),
(70016, 20014, 60016, 'Credit Card', 'Paid'),
(70017, 20015, 60017, 'Afterpay', 'Pending'),
(70018, 20016, 60018, 'Debit Card', 'Paid'),
(70019, 20017, 60019, 'PayPal', 'Paid'),
(70020, 20018, 60020, 'Credit Card', 'Paid'),
(70021, 20019, 60021, 'Afterpay', 'Paid'),
(70022, 20020, 60022, 'Credit Card', 'Paid'),
(70023, 20021, 60023, 'Debit Card', 'Paid'),
(70024, 20022, 60024, 'PayPal', 'Failed'),
(70025, 20023, 60025, 'Credit Card', 'Paid'),

(70026, 20024, 60026, 'Debit Card', 'Paid'),
(70027, 20025, 60027, 'Afterpay', 'Paid'),
(70028, 20003, 60028, 'Credit Card', 'Paid'),
(70029, 20010, 60029, 'Credit Card', 'Failed'),
(70030, 20010, 60029, 'Debit Card', 'Paid');

Part B – Business Analytics Report

1.7 Business concern 1

Name and Description: Platform Revenue Performance

Which social media platform is driving the highest total gross sales during the campaign period?

SQL Query:

```
SELECT      PL.Platform_Name,      ROUND(SUM(OP.Product_Quantity      *
OP.Product_PriceAtCheckout), 2) AS Total_Revenue

FROM PLATFORM PL

JOIN ADVERTISEMENT AD ON PL.Platform_ID = AD.Platform_ID

JOIN SHOP_ORDER SO ON AD.Ad_ID = SO.Ad_ID

JOIN ORDER_PRODUCT OP ON SO.Order_ID = OP.Order_ID

GROUP BY PL.Platform_Name

ORDER BY Total_Revenue DESC;
```

SQL Result:

Platform_Name	Total_Revenue
YouTube	1216.89
Facebook	1106.80
TikTok	1073.99
Instagram	794.99
X	596.00

Source: Author’s analysis

Justification and Insights

The query calculates total gross sales for each platform by multiplying the number of products sold by the product price at checkout.

It joins PLATFORM, ADVERTISEMENT, SHOP_ORDER and ORDER_PRODUCT because PLATFORM contains the social media name, and ADVERTISEMENT relates each ad to a platform using Platform_ID, SHOP_ORDER relates orders to ads using Ad_ID, and ORDER_PRODUCT contains the quantity and price of each item ordered.

Joins are required because revenue is not stored on PLATFORM, therefore we have to trace it from items sold, to orders, to ads, then to platforms.

The query calculates total revenue by platform with SUM() and it rounds to 2 decimal places with ROUND(..., 2), which is typical for currency, thus it groups by PL.Platform_Name to aggregate revenue by platform. It uses GROUP BY PL.Platform_Name so that revenue is calculated separately for each platform.

ORDER BY Total_Revenue DESC is to rank platforms in order of highest to lowest revenue. The query is appropriate because it attributes sales to platforms that drive the related ads, and it computes total gross sales by platform.

According to the query results, YouTube is the largest gross sales driver of the campaign, with the highest revenue at \$1,216.89, meaning its video-centric format is most effective at moving inventory and driving higher-value purchases. Facebook (\$1,106.80) and TikTok (\$1,073.99) are very close to each other in the middle tier, suggesting both traditional social networking and short-form video are consistently driving sales. Meanwhile, Instagram (\$794.99) and X (\$596.00) are far behind, indicating ads on these platforms are currently generating much lower overall sales volumes. However, the figures only take into account gross revenue and don't include platform commission fees required to drive those sales.

These findings suggest David Jones should focus the campaign budget on YouTube, Facebook, and TikTok as these platforms generate significant sales. The team should analyse the ads and audience targeting on Instagram and X to understand the underlying causes that they are not generating as much revenue. However, before shifting the budget, they should conduct another test that takes into account

commission fees, therefore they will have a more comprehensive understanding of the financial implications. The current findings only tell about total sales, thus a platform with high sales might also incur high expenses. David Jones needs to determine the profitability of each platform, so they can make informed decisions about their advertising strategy. Only then will they know what works and what does not, as they will have a clearer understanding of the strengths and weaknesses of each platform.

1.8 Business concern 2

Name and Description: Return on Ad Spend (ROAS)

What is the Return on Ad Spend for each platform (calculated as Total Revenue from Ad divided by Total Commission Cost)? This question is important for David Jones because a platform with high revenue is not always the most cost-effective. ROAS helps compare how efficiently each platform converts advertising commission cost into revenue.

SQL Query

```
SELECT PL.Platform_Name,  
  
       ROUND(SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout), 2) AS  
Total_Revenue,  
  
       ROUND(SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout *  
SO.Order_CommissionRate), 2) AS Total_Cost,  
  
       ROUND((SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout) /  
SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout *  
SO.Order_CommissionRate)), 2) AS Return_On_Ad_Spend  
  
FROM PLATFORM PL  
  
JOIN ADVERTISEMENT AD ON PL.Platform_ID = AD.Platform_ID  
  
JOIN SHOP_ORDER SO ON AD.Ad_ID = SO.Ad_ID  
  
JOIN ORDER_PRODUCT OP ON SO.Order_ID = OP.Order_ID
```

GROUP BY PL.Platform_Name

ORDER BY Return_On_Ad_Spend DESC;

SQL Result

Platform_Name	Total_Revenue	Total_Cost	Return_On_Ad_Spend
X	596.00	26.82	22.22
Facebook	1106.80	57.44	19.27
Instagram	794.99	43.72	18.18
TikTok	1073.99	64.44	16.67
YouTube	1216.89	76.40	15.93

Source: Author's analysis

Justification and Insights

The query calculates ROAS by using total revenue divided by total commission cost. Total revenue is calculated by multiplying the number of products sold by the product price at checkout, while total commission cost is calculated by multiplying the revenue by the commission rate. ROUND(..., 2) is used to format the results to two decimal places.

The query joins the PLATFORM, ADVERTISEMENT, SHOP_ORDER, and ORDER_PRODUCT tables. These joins are necessary because platform performance must be traced from the platform, to the advertisement, to the order, and then to the products purchased.

The PLATFORM table provides the platform name. The ADVERTISEMENT table links each advertisement to a platform using Platform_ID. The SHOP_ORDER table provides the order details and Order_CommissionRate, which links to the ADVERTISEMENT table via Ad_ID. The ORDER_PRODUCT table connects with the SHOP_ORDER table via Order_ID, providing the product quantity and checkout price needed to calculate revenue.

The results are grouped by PL.Platform_Name so each platform is calculated separately, and ordered by Return_On_Ad_Spend DESC so the most cost-effective platforms appear first.

The results of the Return On Ad Spend (ROAS) analysis clearly show an inverse relationship of total sales volume and advertising effectiveness within the channels. Platform X represents the most cost-effective, it has the highest ROAS (22.22), however, it generates the lowest in total sales (\$596.00). Facebook has proven itself to be a well-rounded channel that achieves both significant total sales (\$1,106.80) and a healthy ROAS (19.27). Instagram falls in the middle of the pack with considerable ROAS (18.18) but also struggles with total sales (\$794.99). On the other hand, TikTok and YouTube generate by far the largest amounts of scale (\$1,073.99 and \$1,216.89, respectively); however they are the least effective as measured by ROAS. TikTok generates a ROAS of 16.67 and YouTube is the worst performing at 15.93 because of much higher commissions.

To maximise profitability, David Jones should develop a tiered budget plan. It is logical to rely upon Facebook to be the leading driver of conversions because it has the greatest combination of scale and cost-effectiveness. The company should cautiously expand its spending on X while still testing whether its ROAS is sustainable at greater volumes. Concerning high sales but costly platforms such as TikTok and YouTube, David Jones will want to focus on optimising costs by refining ad targeting strategies that will increase conversion rates or use higher margin products to mitigate the lower ROAS. Since Instagram does not have either effectiveness and substantial volume, David Jones needs to determine how it can play a meaningful and profitable part of the sales process by experimenting with different ad formats (e.g., switching ad spend between Feeds and Reels).

1.9 Business concern 3

Name and Description: Top Performing Ad Campaigns

Identify the top 3 specific advertisements that resulted in the most revenue. This helps David Jones understand which individual ad campaigns were most effective in driving customer purchases.

SQL Query:

```
SELECT      AD.Ad_Name,      ROUND(SUM(OP.Product_Quantity      *  
OP.Product_PriceAtCheckout), 2) AS Ad_Revenue
```

```
FROM ADVERTISEMENT AD

JOIN SHOP_ORDER SO ON AD.Ad_ID = SO.Ad_ID

JOIN ORDER_PRODUCT OP ON SO.Order_ID = OP.Order_ID

GROUP BY AD.Ad_Name

ORDER BY Ad_Revenue DESC

LIMIT 3;
```

SQL Result

Ad_Name	Ad_Revenue
David Jones EOFY Final Days TikTok Ad	746.00
David Jones YouTube Technology Deals	578.99
David Jones Homewares EOFY Facebook Ad	529.00

Source: Author's analysis

Justification and Insights

The query computes the ad revenue by multiplying the quantity of each product sold by its checkout price. It joins three tables: ADVERTISEMENT, SHOP_ORDER, and ORDER_PRODUCT. The ad name is stored in ADVERTISEMENT, customer orders are recorded in SHOP_ORDER, and the products purchased in each order are detailed in ORDER_PRODUCT. ADVERTISEMENT contains the ad name, so SHOP_ORDER connects each order to the ad that triggered it via Ad_ID, and ORDER_PRODUCT contains the quantity and price information needed to calculate revenue, which connects to SHOP_ORDER via Order_ID. The query sums the total revenue per advertisement with SUM() and rounds the result to 2 decimal places with ROUND(..., 2), then GROUP BY AD.Ad_Name separates revenue by advertisement. ORDER BY Ad_Revenue DESC formats the advertisements in descending order based on highest to lowest revenue, and LIMIT 3 restricts the output to the top 3 ads.

Considering individual ad performance, "David Jones EOFY Final Days TikTok Ad" is the top-grossing creative asset in the campaign with \$746.00 in revenue. Although

the previous analysis suggested YouTube is the overall strongest platform, when it comes to individual ads, a TikTok execution is leading the charge. The "David Jones YouTube Technology Deals" ad comes in second place with \$578.99 in revenue, and the "David Jones Homewares EOFY Facebook Ad" is third place at \$529.00. This suggests that urgent, time-sensitive messaging, such as "Final Days", is most effective for driving fast conversions on short-form, trend-focused platforms like TikTok, while more category-focused offers, like Technology and Homewares, continue to drive consistent, steady revenue on more established platforms.

Following these results, David Jones needs to quickly take advantage of the "Final Days" messaging model. The urgent nature of this creative has driven such an exceptional performance through TikTok, therefore it needs to be reviewed and tested on other social media channels including YouTube and Facebook with the goal of replicating this growth in sales. While the particular TikTok ad was the main sales driver, both the YouTube Technology and Facebook Homewares ads have been performing well. David Jones therefore should continue funding these three leading ads; however, they also need to be closely monitoring the frequency to prevent user fatigue prior to completion of the campaign.

1.10 Business concern 4

Name and Description: Category Popularity by Platform

Which product categories sold the most units via each social media platform? This question is useful for David Jones because it shows which product categories perform best on each platform during the campaign.

SQL Query:

SELECT

PL.Platform_Name,

P.Product_Category,

SUM(OP.Product_Quantity) AS Total_Quantity_Sold

FROM PLATFORM PL

JOIN ADVERTISEMENT AD ON PL.Platform_ID = AD.Platform_ID

JOIN SHOP_ORDER SO ON AD.Ad_ID = SO.Ad_ID

JOIN ORDER_PRODUCT OP ON SO.Order_ID = OP.Order_ID

JOIN PRODUCT P ON OP.Product_SKU = P.Product_SKU

GROUP BY PL.Platform_Name, P.Product_Category

ORDER BY Total_Quantity_Sold DESC;

SQL Result:

Platform_Name	Product_Category	Total_Quantity_Sold
Facebook	Women Fashion	4
TikTok	Men Fashion	3
X	Kitchen	3
Facebook	Accessories	2
TikTok	Beauty	2
Instagram	Beauty	2
Instagram	Home	2
YouTube	Technology	2
YouTube	Kids	2
Facebook	Electrical	1
TikTok	Footwear	1
TikTok	Kitchen	1
Instagram	Accessories	1
Instagram	Toys	1
X	Accessories	1
YouTube	Fragrance	1
YouTube	Footwear	1

Source: Author's analysis

Justification and Insights

The query sums the number of units sold for each product category for each platform by using SUM(OP.Product_Quantity). It joins the PLATFORM, ADVERTISEMENT, SHOP_ORDER, ORDER_PRODUCT, and PRODUCT tables. This is because

category popularity can be traced from the platform to the ad, to the order, to the products on the order, and finally to the product's category. PLATFORM contains the platform name and ADVERTISEMENT associates ads with their platform through Platform_ID. SHOP_ORDER links orders with ads through Ad_ID, while also connecting with ORDER_PRODUCT and PRODUCT via Order_ID and Product_SKU, respectively. The former table shows the number of units of each item that were purchased while the latter table indicates the product category of those items through Product_Category.

The results are grouped by PL.Platform_Name and P.Product_Category to separately calculate the total number of units sold for each combination of platform and category. The query sorts the output by Total_Quantity_Sold in descending order, and this allows us to see the top-performing categories on each platform at the top of the list.

Category popularity gives insight into how shoppers leverage each channel differently, and Facebook's Women's Fashion performance is particularly noteworthy with 4 units sold in one category, a very strong fit for apparel on the platform. TikTok also excels in clothing and personal care, leading Men's Fashion with 3 units, and sharing the lead in Beauty with 2 units, while X's strength in Kitchen is also noteworthy with 3 units sold, an important portion of its overall volume. In contrast, YouTube's sales are more thinly spread across several categories including Technology, Kids, and Footwear, while Instagram manages a more balanced albeit lower volume mix of Beauty and Home goods.

Based on the results, David Jones needs to create targeted advertising campaigns for each platform according to their highest-selling categories. For Facebook, Women's Fashion should be the focus of advertising efforts on this platform. Regarding TikTok, more emphasis on Men's Fashion and Beauty products should be placed on this platform's advertising as this is where they perform well. For X, David Jones' team can experiment with more homeware and cooking ads since it responds well to Kitchen products. Considering YouTube and Instagram, Technology and Beauty products perform well on these platforms, therefore more budget should be allocated towards advertising these categories so they can reach their full potential.

1.11 Business concern 5

Name and Description: Platform Performance by Time Window

Which platforms lead to the most orders placed and most revenue generated during 4 time windows: Late night (12AM - 6AM), morning (6AM - 12PM), afternoon (12PM - 6PM), and evening (6PM - 12AM). This is useful for David Jones because the campaign can be improved by scheduling advertisements during the time windows when each platform performs best.

SQL Query:

```
SELECT

    CASE

        WHEN HOUR(SO.Order_Date) BETWEEN 0 AND 5 THEN '1. Late Night
(00-06)'

        WHEN HOUR(SO.Order_Date) BETWEEN 6 AND 11 THEN '2. Morning
(06-12)'

        WHEN HOUR(SO.Order_Date) BETWEEN 12 AND 17 THEN '3. Afternoon
(12-18)'

        WHEN HOUR(SO.Order_Date) BETWEEN 18 AND 23 THEN '4. Evening
(18-00)'

    END AS Time_Window,

    PL.Platform_Name,

    COUNT(DISTINCT SO.Order_ID) AS Total_Orders,

    ROUND(SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout), 2) AS
Gross_Revenue

FROM PLATFORM PL

JOIN ADVERTISEMENT AD ON PL.Platform_ID = AD.Platform_ID

JOIN SHOP_ORDER SO ON AD.Ad_ID = SO.Ad_ID
```

JOIN ORDER_PRODUCT OP ON SO.Order_ID = OP.Order_ID

GROUP BY Time_Window, PL.Platform_Name

ORDER BY Time_Window, Gross_Revenue DESC;

SQL Result:

Time_Window	Platform_Name	Total_Orders	Gross_Revenue
1. Late Night (00-06)	TikTok	1	249.00
1. Late Night (00-06)	YouTube	1	179.99
2. Morning (06-12)	Facebook	3	896.95
3. Afternoon (12-18)	Instagram	4	794.99
3. Afternoon (12-18)	X	3	596.00
3. Afternoon (12-18)	YouTube	1	399.00
4. Evening (18-00)	TikTok	3	824.99
4. Evening (18-00)	YouTube	3	637.90
4. Evening (18-00)	Facebook	1	209.85

Source: Author's analysis

Justification and Insights

The query uses a CASE statement with HOUR(SO.Order_Date) to divide orders into four time windows:

- Late Night: 12AM–6AM
- Morning: 6AM–12PM
- Afternoon: 12PM–6PM
- Evening: 6PM–12AM

The query joins PLATFORM, ADVERTISEMENT, SHOP_ORDER, and ORDER_PRODUCT because platform performance must be traced from the social media platform, to the advertisement, to the customer order, and finally to the products purchased.

COUNT(DISTINCT SO.Order_ID) is used to count the number of unique orders for each platform in each time window. The DISTINCT is important because one order may contain multiple products, and without DISTINCT, the same order could be counted more than once. SUM(OP.Product_Quantity * OP.Product_PriceAtCheckout)

is used to calculate gross revenue, as it multiplies the quantity purchased by the checkout price. ROUND(..., 2) is used to format revenue to two decimal places.

The query groups by both Time_Window and PL.Platform_Name so that order and revenue performance can be compared for each platform within each time period. It then orders the results by time window and gross revenue in descending order, making it easy to identify the strongest platform in each time period.

The results reveal distinct time-of-day purchasing behaviors across the different social platforms. In the morning, Facebook is the clear winner, bringing in a healthy \$896.95 revenue from 3 orders, while in the afternoon (12:00-18:00) total order volume is at its highest with Instagram (4 orders) and X (3 orders) leading the charge, suggesting people are extremely active on these visually driven and text-based platforms during the middle of the day. In the evening, spending shifts heavily toward video platforms, with TikTok and YouTube together bringing in more than \$1,460 gross revenue, and in the late hours of the night, things slow right down with only a few isolated purchases coming through on TikTok and YouTube.

Based on the above patterns, David Jones should be dayparting (time based ad scheduling) to wring additional value out of the advertising budget. Facebook ad spending should be loaded towards mornings where those high value early shoppers are most active. Instagram and X should be ramped up in the afternoon, capitalising on the mid-day browsing and buying urges, while TikTok and YouTube should then receive their share of their investment in the evening, where video content is driving engagement and conversion. However, spending across platforms overnight should be reviewed because the audience is largely inactive and impressions are less likely to convert into sales.

1.12 Business concern 6

Name and Description: Ad-Driven Order Rate

What is the rate of orders successfully linked to an advertisement versus the total number of orders? This question helps David Jones see how much of the campaign's sales activity can be attributed to social media advertising.

SQL Query:

```
SELECT

    COUNT(SO.Order_ID) AS Total_Orders,

    COUNT(AD.Ad_ID) AS Ad_Driven_Orders,

    ROUND((COUNT(AD.Ad_ID) / COUNT(SO.Order_ID)) * 100, 2) AS
    Ad_Driven_Order_Rate

FROM SHOP_ORDER SO

LEFT JOIN ADVERTISEMENT AD ON SO.Ad_ID = AD.Ad_ID;
```

SQL Result:

Total_Orders	Ad_Driven_Orders	Ad_Driven_Order_Rate
29	20	68.97

Source: Author's analysis

Justification and Insights:

The query uses the SHOP_ORDER table because it contains all customer orders. It also joins the ADVERTISEMENT table because advertisements are linked to orders through Ad_ID.

A LEFT JOIN is used so that all orders are included, even if they are not linked to an advertisement. This is important because the query compares ad-driven orders against total orders.

COUNT(SO.Order_ID) counts the total number of orders. COUNT(AD.Ad_ID) counts only orders linked to an advertisement, because null advertisement values are not counted.

The ad-driven order rate is calculated as the percentage of the number of ad-driven orders divided by the total number of orders. ROUND(..., 2) is used to show the percentage to two decimal places.

The results show that out of 29 total orders, 20 of them are attributed to ads, which gives an ad-driven order rate of 68.97%. Hence, paid campaigns are the dominant force behind sales with about 7 out of every 10 checkouts. Only around 31% of orders are coming in organically, meaning people visiting the site directly, through search engines, or returning customers who did not click an ad before making a purchase.

Based on the results, David Jones should continue investing in social media advertising because a large proportion of orders were linked to advertisements. This suggests that the ad campaigns were effective in encouraging customers to place orders. At the same time, David Jones should investigate the orders that were not ad-driven to understand whether they came from organic traffic, direct visits, email campaigns, or other marketing channels. This would help the business better understand the full customer journey.

1.13 Conclusion

In summary, this analysis shows that social media advertising is the primary driver of revenue for David Jones' e-commerce operations, with paid advertising behind 7 out of 10 checkouts. The data highlights that a generic, 'one-size-fits-all' approach to social advertising will not drive strong performance, with activity optimised by platform required. Overall, Facebook is the most reliable all-round channel because it offers a good balance between high sales volume and cost efficiency. Facebook is particularly effective at driving Women's Fashion sales, especially in the morning. Video-led channels, such as YouTube and TikTok, drive the most gross revenue, especially for urgent, trend-based messaging. TikTok is the strongest channel for Men's Fashion and Beauty, while YouTube is the best performing in Technology. Both of them are most effective in the evening, but need careful cost management because they offer relatively low ROAS. In contrast, X offers excellent cost efficiency but in small sales volume. It performs surprisingly well in the Kitchen category, where its most effective time window is the afternoon. The same time window applies to Instagram, although Instagram underperforms the others in terms of total sales.

To create long-term profit and sustainable growth, David Jones needs to shift towards a more targeted marketing strategy that leverages the short-term strength of

paid social, but also places a much greater emphasis on organic channels.

Facebook needs to be positioned as the main conversion workhorse, given its strong combination of volume and cost efficiency, with X scaled up cautiously to maximise its cost-efficiency. Ad spend should be optimised across more expensive video-first platforms such as YouTube and TikTok, enabling the company to achieve a better return on investment. Ad scheduling needs to be highly granular, therefore, Women's Fashion needs to be prioritised on Facebook during the high-converting morning window, whereas Kitchen and Beauty campaigns are run on X and Instagram in the afternoon. Meanwhile, Technology and Men's Fashion campaigns are emphasised on YouTube and TikTok in the evening, when engagement is highest. Moreover, the very clear-cut "Final Days" urgent message from TikTok needs to be rapidly repurposed and tested across all other platforms to see if it can replicate the strong conversion performance it has seen on TikTok. While a paid media strategy refresh is important, given that paid media currently drives almost 70% of all sales, David Jones needs to invest more in organic channels such as SEO, email and loyalty programmes, because this will allow the retailer to reduce its reliance on paid media over time and build a more sustainable and cost-effective customer acquisition engine for the future.